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(54) **MULTIFUNCTIONAL PHOTOCATALYTIC PAINT COAT AND METHOD OF PREPARATION THEREOF**

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See application file for complete search history.

(56) **References Cited**

FOREIGN PATENT DOCUMENTS

CN	1405236 A		3/2003
CN	1696228	*	11/2005
CN	1912031 A		2/2007
CZ	2006829 A3		7/2008
DE	203 06 431 U1		9/2004
WO	WO 2005/044446 A1		5/2005

OTHER PUBLICATIONS

Czech Republic Search Report dated Nov. 11, 2008.
International Search Report (PCT/ISA/210) for PCT/CZ2008/000146 dated Jun. 24, 2009.
Written Opinion (PCT/ISA/237) for PCT/CZ2008/000146 dated Jun. 24, 2009 and Informal Comments dated Aug. 13, 2009.

* cited by examiner

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(57) **ABSTRACT**

The multifunctional paint is based on a highly porous inorganic substance created by a reaction of at least two components. TiO_2 nanoparticles are attached to the surface of this substance. The first component is a water insoluble calcium compound and the second component is a water soluble sulfate. The method of applying the multifunctional paint on the surface is that the first component containing the water suspension of the insoluble calcium compound is applied on the treated area first and subsequently a mixture of TiO_2 nanoparticles suspended in the water solution of the second component is applied over the first layer. Another way is to apply a water suspension of the first component containing also TiO_2 nanoparticles on the treated surface and then deposit the water solution of the second component on the layer. A mixture of all components can be also applied on the treated area at once.

2 Claims, 3 Drawing Sheets

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